

The Dot Product and Orthogonal Vectors

The Dot Product

Defn. The **dot product** (or **inner product**) of two vectors u and v denoted $u \cdot v$ is the sum of the product of corresponding entries

That is

$$u \cdot v = \sum_i u_i v_i.$$

Example Dot Product

For example,

$$\begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 4 \\ 7 \end{bmatrix} = 12 - 4 - 14 = -6.$$

If we view the two vectors as matrices, then the dot product $u \cdot v$ is the entry in the 1×1 matrix given by $u^T v$.

Dot Product is Well Behaved

Several facts one could write. These include the ***distributive law***:

$$u \cdot (v + w) = u \cdot v + u \cdot w$$

Dot Product and Orthogonal Vectors

Defn. Two vectors are **orthogonal** if their dot product is zero.

Orthogonal vectors are sometimes called **perpendicular** vectors.

Norm of a Vector

Defn. The **length** (or **norm**) of vector v is $\|v\| = \sqrt{v \cdot v}$. A **unit vector** has length 1.

Normalization

ALGOR *To obtain a unit vector in the same direction, divide by the length.*

For example, the vector $(3, -1, 2)$ has norm $\sqrt{14}$; a unit vector in the same direction is $\frac{1}{\sqrt{14}}(3, -1, 2)$.

Pythagoras' Theorem

Fact. *Pythagoras' Theorem*

Vectors u and v are orthogonal if and only if
$$||u + v||^2 = ||u||^2 + ||v||^2.$$

Proof is by computation.

Summary

The dot product u and v of two vectors $u \cdot v$ is the sum of the product of corresponding entries. As matrices, the dot product is (the entry in) the matrix $u^T v$.

Two vectors are orthogonal if their dot product is zero. The length/norm of vector v is $\sqrt{v \cdot v}$. To obtain a unit vector in the same direction, divide by the length.

Pythagoras' Theorem is that u and v are orthogonal if and only if $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.